

CO4_AT2 – INDUSTRY PROBLEM SOLVING TASK

1. Autonomous Vehicle Vision System Selection

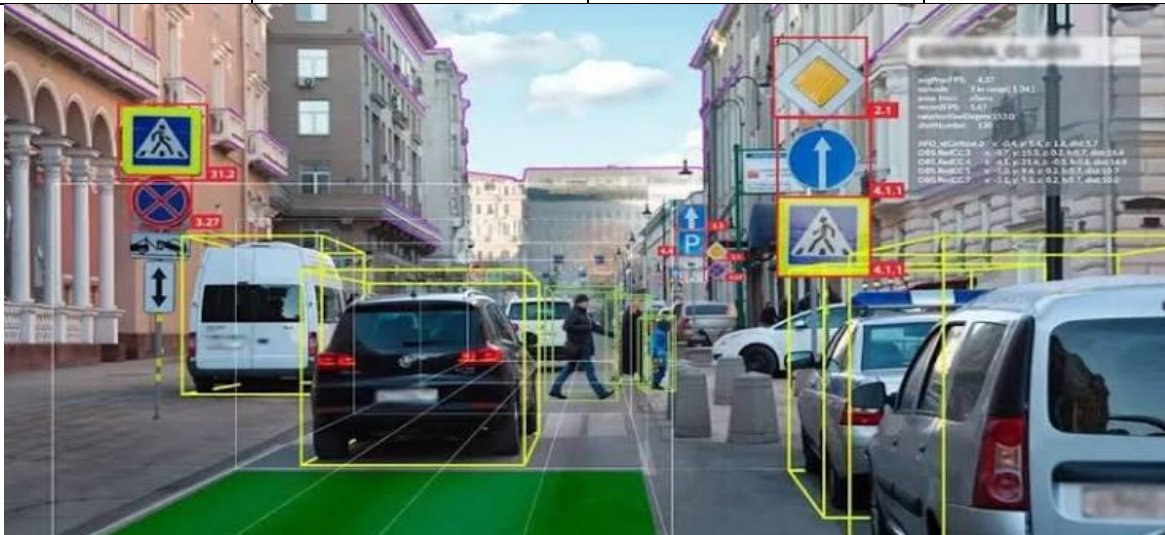
An automobile company is developing an autonomous vehicle detection system. The system must satisfy three conditions: accuracy must not be below 90%, latency must not be above 70 ms, and the vehicle has moderate computational resources. The available methods are Viola-Jones, YOLO and Deep Learning. The correct solution must satisfy all three constraints simultaneously, not just one of them.

Given Conditions

Constraint	Required Condition
Accuracy	$\geq 90\%$
Latency	≤ 70 ms
Computation	Moderate resources available

Available Methods

Method	Accuracy	Latency	Computation
Viola-Jones	82%	30 ms	Low
YOLO	93%	60 ms	Moderate
Deep Learning	97%	90 ms	High



Solution

Step 1: Check Viola-Jones

Accuracy = 82%

$82\% < 90\% \rightarrow \text{FAIL}$ ❌

Latency = 30 ms

$30 \text{ ms} \leq 70 \text{ ms} \rightarrow \text{PASS}$ ✅

Computation = Low $\rightarrow \text{PASS}$ ✅

Decision: Reject Viola-Jones because the required accuracy is not achieved.

↓

Step 2: Check YOLO

Accuracy = 93%

$93\% \geq 90\% \rightarrow \text{PASS}$ ✓

Latency = 60 ms

$60 \text{ ms} \leq 70 \text{ ms} \rightarrow \text{PASS}$ ✓

Computation = Moderate $\rightarrow \text{PASS}$ ✓

Decision: YOLO satisfies all conditions.

↓

Step 3: Check Deep Learning

Accuracy = 97%

$97\% \geq 90\% \rightarrow \text{PASS}$ ✓

Latency = 90 ms

$90 \text{ ms} > 70 \text{ ms} \rightarrow \text{FAIL}$ ✗

Computation = High $\rightarrow \text{FAIL}$ ✗

Decision: Reject Deep Learning because latency and computation conditions are not satisfied.

Final Decision

Viola-Jones $\rightarrow$ ✗ Reject

↓

YOLO $\rightarrow$ ✓ Select

↓

Deep Learning $\rightarrow$ ✗ Reject

Final Answer

YOLO is selected.

Reason: YOLO satisfies all three requirements:

$93\% \geq 90\%$ ✓

$60 \text{ ms} \leq 70 \text{ ms}$ ✓

Moderate computation ✓

Constraint-by-Constraint Evaluation

Method	Accuracy $\geq 90\%$?	Latency ≤ 70 ms?	Moderate Computation?	Decision
Viola-Jones	No (82%)	Yes (30 ms)	Yes/Lower	Reject
YOLO	Yes (93%)	Yes (60 ms)	Yes	Select
Deep Learning	Yes (97%)	No (90 ms)	No (High)	Reject

Logical Reasoning

- Viola-Jones fails because 82% accuracy is below the mandatory 90% threshold.
- Deep Learning fails because 90 ms latency is above the 70 ms limit and its computation requirement is high.
- YOLO satisfies all three conditions simultaneously: 93% accuracy, 60 ms latency and moderate computation.
- The decision must be based on all constraints together; the highest accuracy alone is not sufficient.

Final Selection: YOLO

YOLO is the correct choice for the autonomous vehicle vision system. It provides 93% accuracy, which is above the required 90%, and 60 ms latency, which is within the 70 ms limit. Most importantly, its moderate computational requirement matches the hardware resources available in the vehicle.

Conclusion

Therefore, YOLO is the only method among the three that satisfies every stated constraint. Viola-Jones is fast and computationally light but fails the accuracy requirement. Deep Learning has the highest accuracy but fails the latency and computational requirements. YOLO provides the required accuracy, acceptable latency and suitable computational demand, making it the most practical solution for deployment.

2. Smart Factory Product Inspection Selection

A manufacturing company wants to deploy a real-time computer vision system for detecting defective products on a production line. The system has three mandatory conditions: accuracy must be at least 95%, processing time must not exceed 50 ms, and the available hardware supports moderate computation. The three methods must therefore be tested against all these constraints.

Given Conditions

Constraint	Required Condition
Accuracy	$\geq 95\%$
Processing time	$\leq 50\text{ ms}$
Hardware	Moderate computation

Available Methods

Method	Accuracy	Processing Time	Computation
Traditional Image Analysis	91%	25 ms	Low
YOLO	96%	45 ms	Moderate
Deep Learning	98%	75 ms	High



Solution

Step 1: Check Traditional Image Analysis

Accuracy = 91%

$91\% < 95\% \rightarrow \text{FAIL}$ ❌

Processing Time = 25 ms

$25 \text{ ms} \leq 50 \text{ ms} \rightarrow \text{PASS}$ ✅

Computation = Low $\rightarrow \text{PASS}$ ✅

Decision: Reject because accuracy is below 95%.

↓

Step 2: Check YOLO

Accuracy = 96%

$96\% \geq 95\% \rightarrow \text{PASS}$ ✅

Processing Time = 45 ms

$45 \text{ ms} \leq 50 \text{ ms} \rightarrow \text{PASS}$ ✅

Computation = Moderate $\rightarrow \text{PASS}$ ✅

Decision: YOLO satisfies all conditions.

↓

Step 3: Check Deep Learning

Accuracy = 98%

$98\% \geq 95\% \rightarrow \text{PASS}$ ✅

Processing Time = 75 ms

$75 \text{ ms} > 50 \text{ ms} \rightarrow \text{FAIL}$ ❌

Computation = High $\rightarrow \text{FAIL}$ ❌

Decision: Reject Deep Learning because processing time and computation conditions are not satisfied.

Final Decision

Traditional Image Analysis $\rightarrow$ ❌ Reject

↓

YOLO $\rightarrow$ ✅ Select

↓

Deep Learning $\rightarrow$ ❌ Reject

Final Answer

YOLO is selected.

Reason: YOLO satisfies all three requirements:

96% $\geq$ 95% 

45 ms $\leq$ 50 ms 

Moderate computation 

Checking Traditional Image Analysis

Traditional Image Analysis is fast at 25 ms and has low computational requirements. These are useful characteristics for a production line. However, its accuracy is only 91%. Since the question states that accuracy below 95% may cause defects to be missed, this method cannot satisfy the required reliability condition and must be rejected.

Checking YOLO

YOLO achieves 96% accuracy, which is above the required 95%. Its processing time is 45 ms, which is below the 50 ms maximum. It requires moderate computation, which matches the available hardware. Therefore, YOLO satisfies all three constraints.

Checking Deep Learning

Deep Learning provides the highest accuracy at 98%, so it satisfies the accuracy requirement. However, its processing time is 75 ms, which is above the allowed 50 ms. It also requires high computation, while the hardware supports only moderate computation. Therefore, Deep Learning cannot be selected even though it has the best accuracy.

Constraint-by-Constraint Evaluation

Method	Accuracy $\geq$ 95%?	Time $\leq$ 50 ms?	Moderate Computation?	Decision
Traditional Image Analysis	No (91%)	Yes (25 ms)	Yes/Lower	Reject
YOLO	Yes (96%)	Yes (45 ms)	Yes	Select
Deep Learning	Yes (98%)	No (75 ms)	No (High)	Reject

Logical Reasoning

- Traditional Image Analysis fails because its 91% accuracy is below the required 95%.
- Deep Learning fails because its 75 ms processing time exceeds the 50 ms limit and its computation requirement is high.
- YOLO satisfies accuracy, processing time and computational requirements at the same time.
- For a production line, the selected method must avoid excessive processing delay while still detecting defects accurately.

Final Selection: YOLO

YOLO is the best solution for the smart factory product inspection system. Its 96% accuracy exceeds the required 95%, its 45 ms processing time is within the 50 ms limit, and its moderate computational requirement matches the available hardware.

Conclusion

Therefore, YOLO is the only method that satisfies all the stated constraints. Traditional Image Analysis is sufficiently fast but not accurate enough. Deep Learning is highly accurate but too slow and computationally demanding for the specified hardware. YOLO provides the required accuracy with acceptable processing time and moderate computation, making it the most suitable real-time solution.